

## One 16-mer spans three partners' breakpoints — and only one of the three is a junction any patient is reported to carry

The three FET-family donors are identical over the ten nucleotides before the breakpoint, and the acceptor exon is the same in all three. Every row is TARGET mRNA, sense; the reagent is the reverse complement of the shaded window.

| target mRNA (sense, 5' to 3') |   |   |   |   |   |   |   |   |   | breakpoint |   |   |   |   |   |   |   |   |   |   |   |   |                                           |
|-------------------------------|---|---|---|---|---|---|---|---|---|------------|---|---|---|---|---|---|---|---|---|---|---|---|-------------------------------------------|
| EWSR1 e12::NR4A3 e3           | A | A | T | G | G | T | T | T | G | A          | T | G | A | T | G | C | C | C | T | G | C | G | EWSR1 · reported in patients at this exon |
| FUS e10::NR4A3 e3             | A | C | T | G | G | T | T | T | G | A          | T | G | A | T | G | C | C | C | T | G | C | G | FUS · no exon-resolved report at all      |
| TAF15 e11::NR4A3 e3           | A | C | T | G | G | T | T | T | G | A          | T | G | A | T | G | C | C | C | T | G | C | G | TAF15 · partner reported, this exon not   |



reagent 5'-GGGCATATCATCAAAC-3' (antisense), 8 donor and 8 acceptor bases either side of the seam

Blue, donor exon; green, NR4A3 acceptor exon; boxed and purple, the one position at which the three donors differ. Shaded box, the window this reagent targets.

Every sequence row above is the TARGET mRNA read 5' to 3', not the oligonucleotide. The reagent is 5'-GGGCATATCATCAAAC-3', the reverse complement of the shaded window (5'-GTTTGTATGATATGCCC-3'). Transcribing the shaded letters orders the sense strand, which is a different molecule with no antisense activity.

The same paralogy that lets one reagent cover three fusions is why these designs are hard to discriminate from the parent transcripts: this reagent's gap-level margin is three junction-unique bases inside the six-nucleotide catalytic gap, the window RNase-H1 cleaves. Coverage is predicted from sequence and has not been measured.

Research use only, not for administration. The bases alone, ordered as unmodified DNA, are a different molecule; order from fusion-junction-aso-sequences.csv.